Leveraging Field Data for the Joint Optimization of Capacity and Availability in Low-Margin Optical Networks

Camille Delezoide, Petros Ramantanis, and Patricia Layec

Abstract—**The analysis of a public performance monitoring dataset reveals that reducing margins necessarily amounts to trade availability off for capacity. After carefully establishing this fact, we propose data-driven dynamic rate adaptation as a pragmatic solution to optimally reclaim margins in deployed networks. Based on the dataset, we show that the overall capacity of the monitored network could have been increased by 106% while maintaining the availability over 99.99% for 95% of the connections. Beyond such promising results, our work provides the fundamental tools to predict - for any network - how capacity upgrades would impact the availability, and to identify the most suitable rate adaptation mechanism to optimize margins.**

Index Terms—**Automation, availability, data mining, design optimization, monitoring, optical fiber networks.**

I. Introduction

THE spectral efficiencies achieved with state-of-the-art network elements are within an order of magnitude of the Shannon limit. Thus, keeping pace with the steady cost-per-bit reduction while achieving the unprecedented levels of reliability required by service providers is becoming more and more challenging. In this context, optical network margins constitute a largely untapped reservoir of network capacity. However, knowing in practice how much margin can be converted into extra capacity remains an open question.

In 2017, a dataset containing 14 months of performance monitoring of a large North-American optical backbone network was released [1]. While the exact network topology is not available, Q^2 values from February 2015 to April 2016 are reported and organized in 4000 time series, one for each monitored connection. Each connection is composed of a single, 100 Gb/s, PDM-QPSK optical channel propagating over a single lightpath, and all lightpaths are shared by multiple channels. Reported Q^2 values are averages over 15-minute-periods.

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In [2], Ghobadi *et al* have leveraged the first three months of the dataset to evaluate the potential capacity gains achievable with rate elasticity. However, the authors did not consider that, due to the major variations of the quality of transmission (QoT) - the *Q-drops* - reported in [3], any permanent rate upgrade could have a negative impact on the availability [4] of the upgraded path and its associated connection. More generally, any low-margin scheme as depicted in [5]–[9] could indirectly lower the availability because of QoT variations.

In [10], we performed a field trial of *marginless* operation in which the rate was dynamically set based on monitored signal-to-noise ratio [11] (SNR) values as QoT metric. Leveraging software defined network (SDN), elastic transponders and data-aided SNR prediction, we demonstrated the robustness of the *marginless* mode to SNR fluctuations, and its ability to adapt to emulated aging without outages. Dynamic rate adaptation thus appeared as an efficient solution to jointly optimize capacity and availability. As follow-up, we leverage here the dataset in [1] to assess the benefits of dynamic rate adaptation in terms of capacity and availability on a full scale network with genuine performance variations.

The paper is organized as follows. In Section II, we analyze the performance data to assess SNR margins and how they vary in time. Then, in Section III, we leverage the same data to quantify the trade off between capacity and availability inherently stemming from SNR variations. Based on the results of Section II and III, we propose and simulate two opposite rate adaptation mechanisms achieving joint optimization of capacity and availability. More specifically, we study in Section IV a memoryless and dynamic (M-D) rate adaptation directly inspired from the *marginless* mode. In contrast, the causal and dynamic (C-D) mode tested in Section V leverages the concept of statistical outlier to statically set the rate after each service interruption. In the conclusion, we sum up all important results and discuss how the methodology presented in this paper may be used to optimize margins in any monitored network.

II. Large Scale Assessment of SNR Margins and Their Time Variations in a Deployed Network

Throughout the paper, we use the SNR as defined in [11] as reference QoT metric. To avoid any ambiguity, we specify that the SNR we consider is measured within the coherent receiver, right before forward error correction (FEC). Note

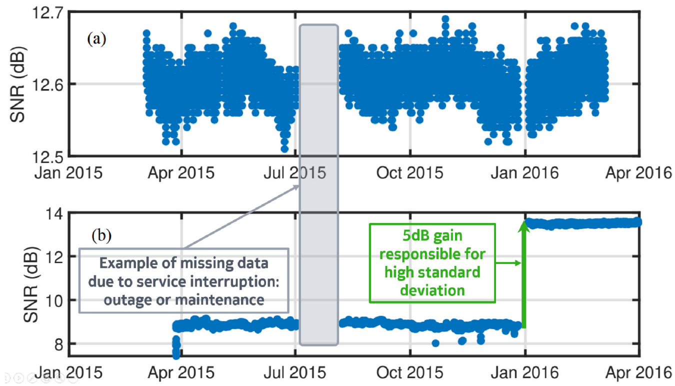

Fig. 1. Monitored SNR for (a) channel 1395 of segment 36 and (b) channel 899 of segment 26, with standard deviation of respectively 0.2 dB and 2.1 dB.

that in the optical communication industry, this SNR is often called *electrical* SNR (ESNR) in opposition to the *optical* SNR (OSNR). The OSNR measures the quality of the optical signal reaching the receiver, initially only accounting for ASE noise. In contrast, the ESNR or simply SNR includes all noises relevant to the detection. As consequence, under the assumptions of the Gaussian Noise (GN) model [12], there are simple and accurate closed-form relations between the SNR, the Q^2, and the pre-FEC bit error ratio (BER) [13]. In particular, the SNR is approximately equal to the Q^2 in PDM-QPSK. Since all monitored connections in the dataset [1] are in PDM-QPSK, we directly convert the Q^2 values reported in the dataset into SNR values.

After processing Q^2 values as SNR values, we convert each SNR into a time-dependent SNR margin M_{tot} using:

$$M_{tot}[dB](t) = SNR[dB](t) - SNR_{FEC}[dB] \qquad (1)$$

In absence of deeper knowledge on the monitored connections, we consider the SNR forward-error correction (FEC) limit for the default 100 Gb/s rate to be 5 dB, as provided in [2]. This value, corresponding to a pre-FEC bit-error ratio (BER) of 3.77e^{-2}, is consistent with the performances of state-of-the-art commercial transponders.

An initial study focused on a few randomly selected connections revealed a great variety of time-dependent behaviors. For example, we plot in Fig. 1 two extremes in terms of standard deviation. The SNR evolution for the connection labeled "channel 1395" plotted in Fig. 1(a) is very stable, with a standard deviation of 0.2 dB. In contrast, the standard deviation of the SNR for the connection labeled "channel 899" plotted in Fig. 1(b) is 2.1 dB. This high deviation is due to a SNR increase of almost 5 dB around January the 1st, 2016. Note that the missing data points, e.g. in July 2015, correspond to either outages or maintenance periods [3], a maintenance being any voluntarily service interruption decided by the network operator.

In the following, we first select a connection - labeled "channel 112" in the dataset - to perform a case study. We choose this specific connection for its representativeness of the major types of SNR variations found throughout the dataset. Later, we apply the teachings of this case study to analyze the whole dataset with suitable statistical tools.

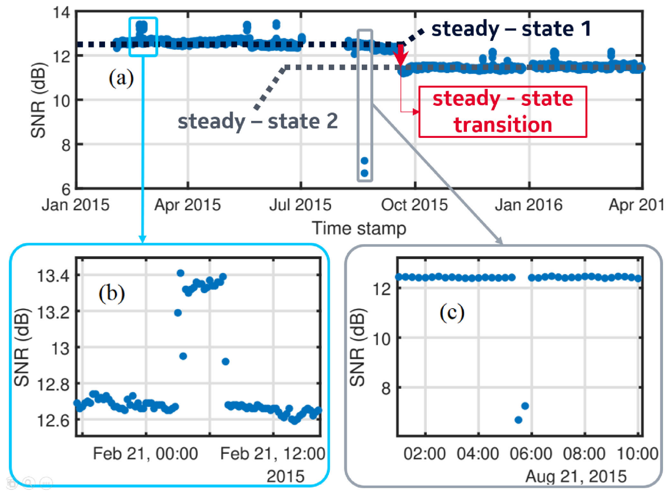

Fig. 2. Monitored SNR for the connection labeled "channel 112," (a) complete time series, (b) zoom on temporary upward variation and (c) zoom on fast downward variation or *Q-drop*.

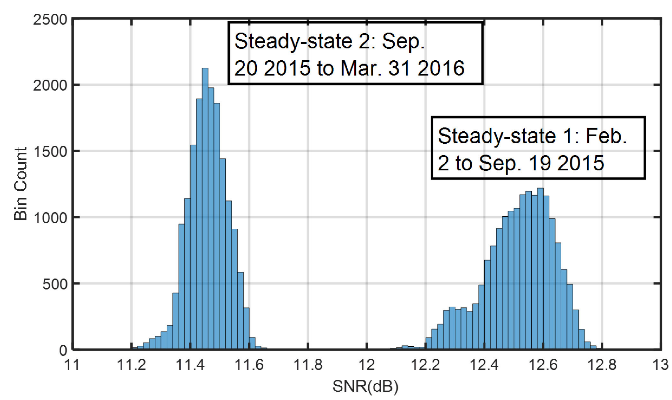

Fig. 3. Histogram of the monitored SNR for the connection labeled "channel 112". Each lump represents a steady-state.

A. Case Study of SNR Variations in "Channel 112"

We plot the SNR as function of time in Fig. 2(a). The curve shows several types of variations throughout the 14-month monitoring period. We can first observe several temporary variations, e.g. in Fig. 2(b), lasting for several hours and with amplitudes around 1 dB.

Secondly, on one occasion on August, 21 2015, the measured SNR temporarily drops by more than 5 dB. Fig. 2(c) zooms in on this event. Since dataset values are averaged over a 15-minute-period [3], the actual SNR variation may have been larger. Such sudden drops of performance have been reported and labeled as *Q-drops* in [3]. Although not explained, authors observed that Q-drops generally announce an outage. This suggests that Q-drops may be due to major physical layer issues which are either very limited in time or rapidly compensated by some automation, but eventually leading in most cases to a full connection breakdown.

Thirdly, we notice in Fig. 2(a) and in the corresponding histogram (Fig. 3) that most of the SNR samples are centered around two steady-state values. Before the transition in September 2015, the SNR is stable around 12.5 dB. After the transition, the SNR

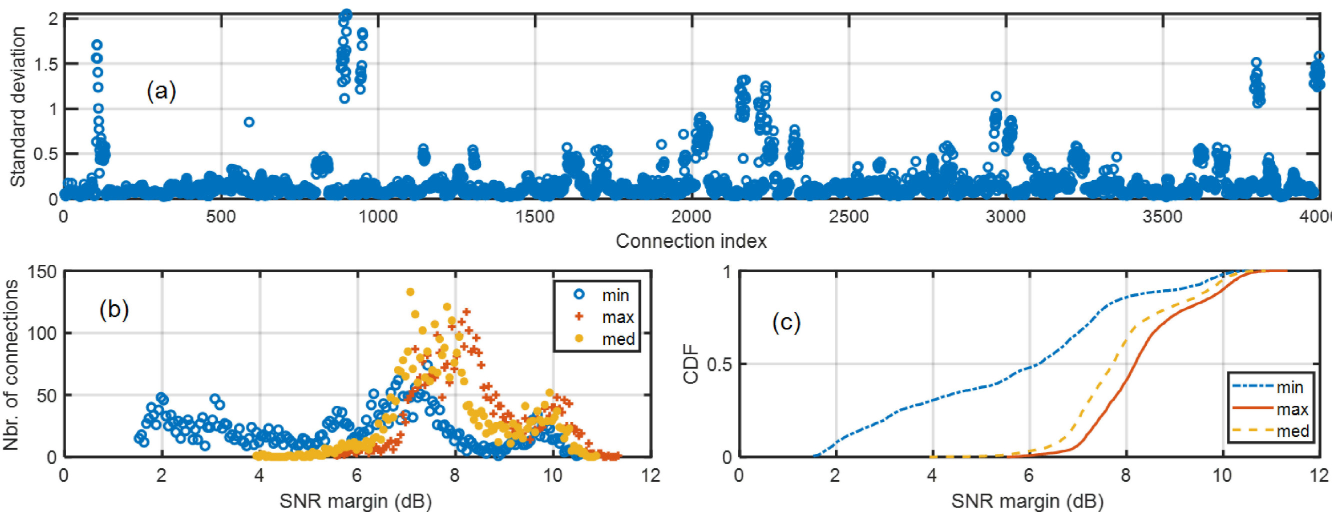

Fig. 4. (a) Standard deviation of the SNR margin calculated for each optical connection, (b) histogram and cumulative distribution function (CDF) (c) of the minimum, maximum and median SNR margin relative to the monitoring period of each connection.

is stable around 11.5 dB. The transition from one steady-state to another can only denote a significant modification of the optical connection. In a deployed network, this can occur for various reasons: fiber cuts and repairs, replaced hardware, removal of dispersion compensation modules (DCMs), defragmentation, removal or addition of channels, etc. Note that depending on the modification, its impact on performance can be either negative as in Fig. 2(a), or positive as in Fig. 1(b).

Finally, most of the spread of the two lumps in Fig. 3 are due to low-amplitude fluctuations with multiple time scales. Their amplitudes are in the 0.1 dB range, with a maximum at 0.3 dB.

A major teaching of the case study is that, generally-speaking, we cannot treat the SNR as normally distributed at any time. Indeed, the distributions associated with steady states, e.g. the lumps in Fig. 3, are not Gaussian-shaped. As a result, we cannot describe SNR time series simply by their averages and standard deviations. More, the presence within steady-states of reversible SNR variations such as Q-drops causes severe and unpredictable imbalances between the edges of each SNR distribution. For a global study of the network, this indicates the need to evaluate for each connection the extreme, i.e. minimum and maximum, and median SNR values, and the long-term variations of steady-state values.

B. Distributions of the Minimal, Maximal and Median Margins for all Connections in the Dataset

In this section, we calculate for each connection i.e. each time series, its minimum, maximum, median SNR values and the SNR standard deviation. For instance, we find that 6.69 dB is the minimum SNR value for "channel 112" studied previously, i.e. the smallest SNR value observed for this connection during the whole duration of the monitoring. Based on these calculations, we plot in Fig. 4 various graphs providing a network-wide overview of the time variations of SNR margins, using (1) for conversion.

In Fig. 4(a), we plot the standard deviation of the SNR margin as a function of the connection, indexed from 1 to 4000. The observation of Fig. 4(a) confirms the wide variety of behaviors illustrated in Fig. 1, although a majority of connections have low variability. A deeper analysis shows that 80% of connections have a standard deviation below 0.2 dB, and 2.5% of connections are over 1 dB, which can be explained by steady-state transitions such as those observed in Fig. 1(b) and Fig. 2(a).

Next, we plot in Fig. 4(b-c) the histogram and cumulative distribution function (CDF) of the calculated minimum, maximum and median SNR margins. We observe that due to Q-drops, there is a significant difference between the distributions of minimum and median SNR margins. In the context of network design, the worst-case approach would consist here in solely considering the margin minimum, for instance to plan for upgrades. However, since minimal values are generally statistical outliers, this would lead to vastly underestimate the potential of the network for optimization. The observation of Fig. 4(b-c) confirms the need to reconsider the worst-case approach, and to evolve towards a statistical approach of network design as developed in [8]. Nevertheless, we observe from Fig. 4(c) that around 50% of optical connections have a minimal SNR margin of 6 dB or more, and 10% have a minimal margin exceeding 9 dB. This underlines the great potential of this network and potentially others for significant capacity optimization, even with conservative approaches derived from the worst-case paradigm.

C. System Margins and Long Term Performance Variations

In known design practices, system margins are associated with how the QoT varies in time. Depending on the sources [14], authors estimate that system margins are typically anywhere between 3 dB and 6 dB. Since commercial 100 G PDM-QPSK transponders are contemporary to the dataset, it is likely that connections have been monitored close to the begin-of-life (BOL). Therefore, the SNR margins observed in the previous section include most of the system margins. System margins thus appear as a dominant contribution to overall margins.

In the rationales for system margins, network designers first account for the progressive addition of channels propagating in the fiber links. This slowly increases the nonlinear noise due to the Kerr effect, thus reduces the SNR of pre-existing connections. Such reductions shall appear after a maintenance period as a downward step, as may be the case in Fig. 2(a). The amplitude of the downward step could easily be estimated using available QoT estimation models [12], provided the model parameters are known.

Then, it is often considered that performances of network elements such as fibers and amplifiers degrade over time. This phenomenon is typically called aging. Although aging values circulate in the literature [14], the origin of such values is - to the best of our knowledge - never disclosed. This lack of reliable aging data can lead to inappropriate margin settings.

On top of such permanent SNR decreases, positive or negative SNR variations can occur for other reasons. As all optical devices, network elements are sensitive to external factors such as temperature [15]. This sensitivity can be related to birefringence effects [16]. Depending on how network elements are deployed, temperature may have more or less influence, e.g. an aerial cable compared to a buried cable, or depending if the storage room is temperature-controlled. Such effects can typically explain the SNR variations observed in Fig. 1(a), with an order of magnitude of 0.1 dB.

Last but not least, Fig. 1(b) shows that the average SNR can significantly increase. Such improvements of the QoT can be explained by very specific maintenance operations, such as the removal of legacy 10 G channels and associated dispersion compensation units in the optical links used by the monitored connections [2], or due to the reallocation of the connections' wavelengths in the context of a defragmentation.

In the following, we leverage the dataset to quantify the cumulated impact of all possible SNR variations on the long term evolution of the SNR, with potential repercussions on how to optimally reclaim system margins.

D. Distribution of Long Term SNR Variations in the Studied Network

Here, we measure for each connection the steady-state value at beginning and end of the time series, respectively SNR_{begin} and SNR_{end}. More precisely, SNR_{begin} and SNR_{end} are the median SNR values over the first month and the last month, respectively. The median is chosen instead of the average for its lower sensitivity to statistical outliers such as Q-drops.

For simplicity, we call *aging* the difference for a given connection between SNR_{end} and SNR_{begin}. Since variations, positive or negative, are more likely to be observed over longer monitoring periods, we consider the *aging coefficient* as the *aging* normalized by the duration of the period over which the *aging* is calculated.

$$\textit{aging coefficient} = \frac{\textit{aging}}{\Delta T_{mon}} = \frac{SNR_{end} - SNR_{begin}}{T_{end} - T_{begin}} \quad (2)$$

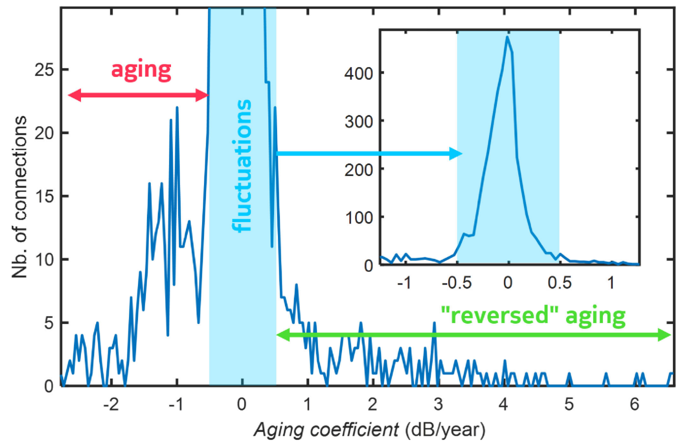

Fig. 5. Distribution of the *aging coefficient*. Inset: zoom on the central lobe.

where $\Delta T_{mon} = T_{end} - T_{begin}$ is total duration of the monitoring. We plot the corresponding histogram in Fig. 5.

We observe that most connections - 85% - have an *aging coefficient* within +/$-$0.5 dB per year. The median value is -0.05 dB/year meaning that roughly 50% of the connections have a positive *aging coefficient*. Left of the histogram, we observe the optical connections for which the cumulative impact of SNR variations results in a negative *aging* value, i.e. *regular aging*. Roughly 10% of monitored connections have an *aging coefficient* below -0.5 dB/year. On the right side, 5% of connections exhibit what can be called *reversed aging*, i.e. an aging coefficient above 0.5 dB/year, i.e. above what can be considered to be due to fluctuations. More, 3% of all connections have an *aging coefficient* above 1 dB/year, 1.6% above 2 dB/year, 0.8% over 3 dB/year and 0.15% over 5 dB/year. Note here that in such instances of *reversed aging*, the measured values should not suggest that a steady increase of the SNR may be observed. Indeed, SNR increases are only in the form of step-like variations after service interruptions, as can be observed in Fig. 1(b), and obtained values result from the indirect averaging in the calculation of *aging coefficients*.

The results show that long term SNR variations are clearly not homogeneous across the monitored network. Also, contrary to what is generally assumed, such variations are not solely downwards. Here, this is only true for half of the connections. Therefore, it clearly appears that a permanently efficient use of network resources can only be achieved by dynamically adapting the transported capacity in a per-connection basis. This is studied in Section IV and V.

III. Assessment of the Trade-Off Between Capacity and Availability

The trade-off between maximal capacity and reliability/availability has previously been underlined by several authors [17], [18], but not quantified. In this section, we leverage the dataset to numerically assess the impact of virtual, static and arbitrary margin reductions on the availability of the connections.

TABLE I
CORRESPONDENCES OF FORMATS, RATES AND ASSOCIATED FEC THRESHOLDS

PDM Format	QPSK	8-QAM	16-QAM	32-QAM	64-QAM
Rate (Gb/s)	100	150	200	250	300
SNR FEC threshold (dB)	5	8.3	11.3	16.8	22.2

A. Method and Definitions

We first evaluate the SNR thresholds corresponding to a pre-FEC BER of $3.77e^{-2}$ [2] as a function of the net rates achievable with a 32GBd PDM coherent optical channel, i.e. 100 Gb/s in QPSK, 150 Gb/s in 8QAM, 200 Gb/s in 16QAM etc using formulas in [13]. The correspondences of formats, rates and SNR FEC thresholds are displayed in Tab. I. We consider the intermediary rates to be achieved by format adaptation, through hybrid modulation formats or probabilistic shaping, and neglecting implementation penalties. For simplicity, we assume that the SNR thresholds for the intermediary rates can be determined using a spline interpolant on the standard formats. Furthermore, we assume that changing the rate does not modify the SNR. This common assumption of the GN model is largely verified in linear or weakly nonlinear propagation regimes in uncompensated links. In other cases, extra nonlinear penalties would have to be applied as in [2], eventually leading to a reduced capacity gain achievable with elasticity. The extra penalties could easily be calculated with the more recent QoT models. In our case however, this would have required a deeper knowledge of the network than what is publicly available.

For each monitored connection, we evaluate the *availability* [4] as a function of the rate setting. The availability A is the ratio between the available time (AT) and the total time (TT). The unavailable time (UT) is simply the difference between total and available times $UT = TT - AT$, so that:

$$A = \frac{AT}{TT} = 1 - \frac{UT}{TT} \tag{3}$$

In our simulations, we use Eq. (3) to calculate the availability of a connection for a given rate setting from the estimation of the corresponding unavailable time, for which there are two separate contributions.

The first contribution are the actual outages that took place during the 14 months of the data collection, with a static rate at 100 Gb/s for all connections. Since these outages are independent from our rate adaptation simulations, we call them *external* outages in the following, and associate them to the *external* unavailable time UT_{ext}, i.e. the unavailable time due to *external* outages. In the dataset, outages are not explicitly reported. However, we know from [3] that missing monitoring data indicates a service interruption, i.e. an outage or a maintenance window. Therefore, we estimated UT_{ext} for each connection by summing up all time periods during which data is missing, a single missing entry corresponding to 15 minutes of unavailable time. Note that this definition of what constitutes an unavailable time differs from the standard according to which maintenance windows should be excluded.

The second contribution to the unavailable time UT are the virtual outages caused by our simulated rate upgrades. In opposition to the natural, *external* outages, we call these latter outages *internal*, and associate them to the *internal* unavailable time UT_{int}. For each rate setting, UT_{int} is estimated by summing up all 15-minute periods for which the average SNR is below the FEC limit corresponding to the simulated rate. The same principle is applied in Section IV and V with a dynamically changing rate.

In the end, we can write:

$$\begin{aligned} A &= \frac{AT}{TT} = 1 - \frac{UT_{ext}}{TT} - \frac{UT_{int}}{TT} = A_{ext} + A_{int} - 1 \\ A_{ext} &= 1 - \frac{UT_{ext}}{TT} \quad \text{and} \quad A_{int} = 1 - \frac{UT_{int}}{TT} \end{aligned} \tag{4}$$

where we introduce A_{ext} and A_{int} as the *external* and *internal* availabilities, respectively. Note that in (4), availabilities are expressed in direct form, i.e. not in percents. By construction, *external* and *internal* outages are independent and cannot occur at the same time. Therefore, the only cause for unavailability at the default 100 Gb/s rate is external, resulting in $UT_{int} = 0$ min and $A_{int} = 100\%$ at 100 Gb/s. Also, contrary to A_{int}, A_{ext} is independent from the simulated rate setting.

Finally, we point out that both the 15-minute-refresh rate and the overall monitoring duration of 14 months limit the availability resolution. For instance, a single 15-minute outage over 14 months leads to A = 99.9975%, i.e. between four and five-nines. Exploring availability values above 99.9975% with the same monitoring duration would require shorter refresh rates, e.g. 6 minutes for five-nines, and 37 seconds for six-nines resolution.

Similarly to the previous section, we begin with a study of a single connection to illustrate the behaviors to be expected throughout the network. Based on the teachings of this case study, we will proceed with an analysis of the whole dataset.

B. Case Study: Impact of a Rate Upgrade on Availability for "Channel 112"

In Fig. 6(a), we plot the monitored SNR with thresholds corresponding to rates from 100 Gb/s to 250 Gb/s by steps of 25 Gb/s. This plot illustrates the various contributions to the unavailability of the connection. For instance, a significant contribution to UT_{ext} is the missing data spanning from July to August 2015. Regarding UT_{int}, let us briefly consider a simulated rate at 150 Gb/s corresponding to a FEC threshold at 8.3 dB (cf. Tab. I). Then, the only *internal* outage in such simulation occurs because of the Q-drop on August, 21 2015 (cf. Fig. 2(c)), resulting in $UT_{int}(R = 150\text{Gb/s}) = 30$ min. For rates higher than 175 Gb/s, the SNR thresholds are much closer to the typical SNR values for this connection, resulting in a fast decay of A_{int}.

For a deeper analysis, we estimate the availability for all rates between 100 Gb/s and 300 Gb/s with a 0.1 Gb/s granularity. Results are plotted in Fig. 6(b) in log scale, i.e. in number of

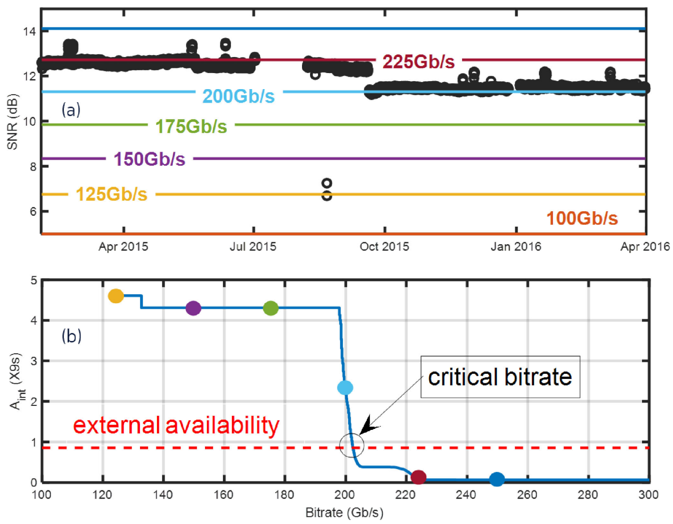

Fig. 6. (a) SNR in time and SNR thresholds corresponding to rates from 100 Gb/s to 250 Gb/s by steps of 25 Gb/s; (b) Estimated availability A_{int} as a function of the rate for channel 112 and comparison with A_{ext} (dashed line) in logarithmic scale, i.e. in number of nines (X9s).

nines. For comparison, we also represent the external availability A_{ext}, equal to 85.9% for the studied connection.

From Fig. 3, we know that the wide majority of the collected SNR values are over 11 dB. This explains why, beyond 100 Gb/s, the internal availability A_{int} remains close to 100% until the rates have SNR thresholds exceeding 11 dB. Also observable in Fig. 6(a), this occurs close to 200 Gb/s. The step-like variation of A_{int} around 130 Gb/s in Fig. 6(b) occurs because of the 15-minute granularity of the virtual unavailable periods. Beyond 200 Gb/s, the space between the two lumps of the SNR distribution (cf. Fig. 3) converts into an availability plateau, i.e. a region where the rate can be increased with little to no impact on the availability. Finally, when the SNR threshold exceeds 13 dB, there is no sufficient SNR value at any time to support the corresponding rates. Consequently, A_{int} tends to zero.

The plot in Fig. 6(b) also allows to compare the rate-dependent internal availability A_{int} to the constant external availability A_{ext}. We can determine a remarkable rate value, R_{crit}, for which $A_{int} = A_{ext}$. It is equal to 202 Gb/s for the studied connection. R_{crit} separates the two fundamental limitations of the connection's availability A, as formally established in (4). For rates lower than R_{crit}, the availability is limited by *external* outages whereas for rates higher than R_{crit}, it is limited by *internal* outages, i.e. the outages occurring when the monitored QoT is too low compared to the FEC threshold corresponding to the simulated rate setting.

C. Setting Margins Based on Availability Estimations

The observations derived from this case study provide first insights on how monitoring data can be leveraged to reduce margins optimally, i.e. with minimized impact on availability. Two approaches may be considered. The first approach would consist in allowing rate upgrades, i.e. margin reductions, as long as the predicted *internal* availability A_{int} remains - for instance - above five nines. In opposition, the second approach would be to consider that the rate can be upgraded as long as the predicted impact on A is negligible.

Whereas the first approach attempts to minimize the probability of *internal* outages, i.e. outages due to the rate elasticity, the second approach puts the same probability in a broader context, by implicitly comparing it with the probability of an *external* outage. This is especially important for networks where the availability spans over several orders of magnitude depending on the connection, as depicted in [3]. From (4), we understand that if for instance A_{ext} is below two nines, the *internal* outages will not significantly lower the overall availability A as long as A_{int} remains above three nines. Indeed, if $A_{ext} = 99\%$ and $A_{int} = 99.999\%$, then $A = 98.999\%$ whereas $A_{int} = 99.9\%$ leads to $A = 98.9\%$ with an unchanged A_{ext} value. In this example, the second approach allows to significantly reduce the constraint on A_{int} compared to the first approach, which can in turn allow much higher rate upgrades. We simulate and compare both approaches in the following.

D. Availability as a Function of the Rate for all Connections

Here, we leverage the dataset to quantify how many connections may be upgraded to a given rate when a constraint on availability is enforced. Both *internal* and *external* availabilities - A_{int} and A_{ext} respectively - are calculated using the method described in III.A. For each of the 4000 connections, we first evaluate A_{ext}, then A_{int} as a function of the simulated rate setting, which varies between 100 Gb/s and 300 Gb/s with 0.1 Gb/s granularity. Finally, we determine for a given *internal* availability target A_{target} the maximum rate $R_{\max}$ for which $A_{int} > A_{target}$ is verified.

For the first margin setting approach discussed in III.C, the target for internal availability is the same for all connections. We test three values for A_{target}: 99.9%, 99.99% and 99.999%. For the second approach, the target is individually set for each connection based on the *external* availability, i.e. $A_{target} = A_{ext}$. Note that for such target, $R_{\max}$ coincides with R_{crit}. From all $R_{\max}$ values obtained, we determine the number of connections compatible with a given rate and availability target. These results are displayed in Fig. 7.

Let us consider the plot corresponding to $A_{target} = 99.999\%$. As it turns out, all connections could withstand a rate at 120 Gb/s while maintaining $A_{int} > 99.999\%$. For larger rates however, the proportion of compatible connections rapidly decreases due to the drastic constraint on availability: they are 60% at 180 Gb/s, and only 20% at 220 Gb/s. In comparison, an availability target set at 99.9% increases to 50% the proportion of connections compatible with 220 Gb/s.

Due to the large number of service interruptions from outages or maintenance operations, our estimations of the *external* availability are generally quite low, from 65 to 98%. Therefore, $A_{int} > A_{ext}$ logically appears as the lesser constraint, achieving the highest proportions of compatible connections for all rates. However, this result is partly due to the fact that we count maintenance operations as unavailable time. A similar study where only outages would contribute to unavailability would

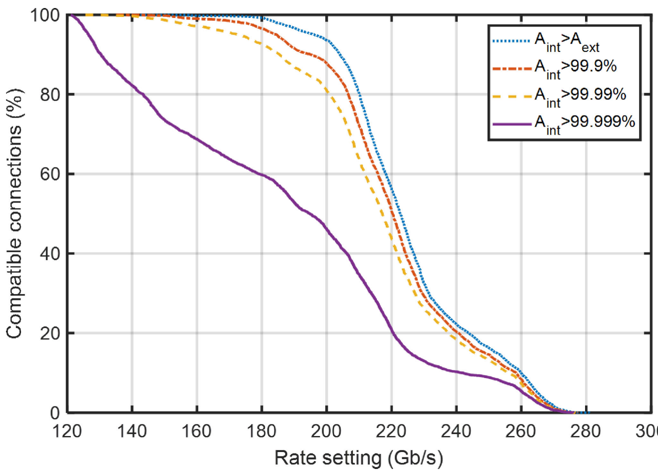

Fig. 7. Proportion of connections compatible with arbitrary rate settings for various availability targets.

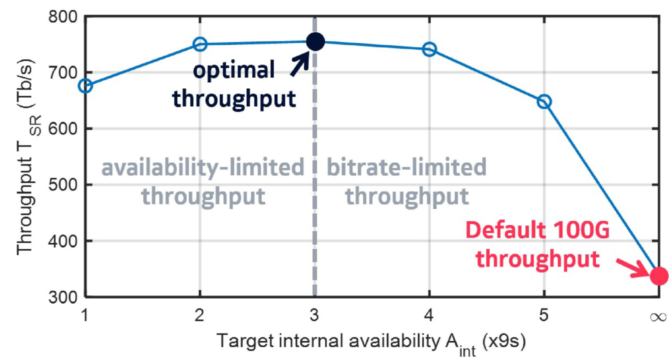

Fig. 8. Network throughput T as a function of the *internal* availability target.

probably produce more qualified results. This illustrates the necessity to add sufficient context to the monitoring datasets to allow proper data analysis.

We finally observe significantly different results between four and five nines target availabilities. Note that a single 15-minute outage per year leads to a 99.997% availability. Therefore, the constraint that $A_{int} > 99.999\%$ roughly means that there should not be a single SNR value below the FEC threshold during the entire duration of the monitoring. Thus, the corresponding plot is extremely sensitive to statistical outlier SNR values associated with Q-drops. Those are found at least once in 50% of the connections. In comparison, a 99.99% availability target allows up to three 15-minute outages per year. Thus, it clearly appears that part of the significant difference between the two availability targets is due to the coarse refresh rate of the monitoring data, coupled to the presence of Q-drops. For availability analysis, this illustrates the need for faster sampling, or alternatively the need for collecting additional metrics such as the number of errored seconds.

E. Margins and Network Throughput

In our simulations, a connection could be set with an arbitrarily high rate leading to 0% availability. Therefore, the estimation of the network throughput in terms of capacity must account for both rate setting and resulting availability.

In the context of virtual optical networks (VONs), authors in [18] introduced the time average aggregated data rate (TAADR) as optimization metric. The TAADR is defined as the sum over all connections k of the product between the connection's rate R_k and the corresponding availability A_k:

$$\text{TAADR} = \sum_{connections} R_k A_k \tag{5}$$

The TAADR is a suitable indicator of the network capacity throughput when the rates of all connections are static. Since we later aim to study the impact of dynamic, data-driven rate adaptation on network throughput, we need to adapt the TAADR metric to be compatible with time-varying rates.

In the static case, the product between rate and availability coincides with the time-average of the capacity delivered by a connection. In the dynamic case, we can consider that the delivered capacity C_k is time-dependent, equal to the rate under normal operation, and zero during outages or maintenance periods. Thus defined, C_k includes the availability and is compatible with a time-dependent rate. Now, we can generalize the TAADR in the form of a network throughput T defined as the sum over all connections of individual connection throughput T_k, i.e. the time-averaged capacity $T_k =< C_k >_t$:

$$T = \sum_{connections} T_k = \sum_{connections} < C_k >_t \tag{6}$$

F. Capacity-Availability Trade-Off and Margin Sweet Spot

Here, we leverage the numerical results of III.D to evaluate the network throughput as a function of the *internal* availability target. Results are plotted in Fig. 8.

With all 4000 monitored connections at 100 Gb/s, the throughput would be 400 Tb/s with 100% availability for all connections. Due to outages and maintenance periods however, the throughput achieved is lower, equal to $T_{\text{ref}} = 340$ Tb/s. This throughput is used in the following as reference for performance comparison.

With rate settings designed to achieve at least 99.999% internal availability for every connection, the simulated throughput is 648 Tb/s, almost double compared to the reference 100 G case. We further observe that the network throughput reaches an optimal of 755 Tb/s for an internal availability target of 99.9%. Elsewhere, the throughput is either limited by low availability or by low rates. We finally observe that the throughput variation between the 99% and 99.99% targets does not exceed 2%.

The bell curve obtained in Fig. 8 unequivocally demonstrates the trade-off between capacity and availability at the network scale, with several important consequences. First, a network should not be operated in the availability-limited domain, left of the throughput optimum, where both availability and throughput are suboptimal. This corresponds to excessively low margins. Depending on the point of view, the optimal margin setting could be when the throughput optimum is achieved, or when a satisfactory middle ground is found between improved throughput and necessarily degraded availability, on the right-side of the

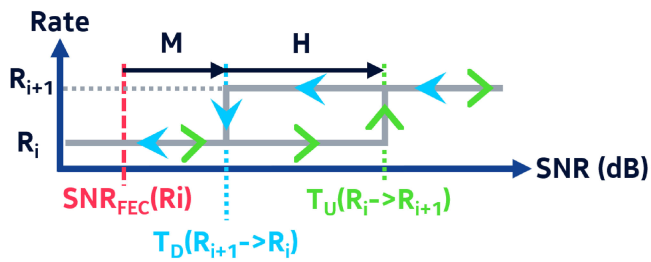

Fig. 9. Network throughput in M-D mode as a function of (a) the hysteresis for M=0.1 dB and (b) the residual margin for H=0.1 dB. CDF of connection internal availability (c) and throughput (d) for G=25 Gb/s, H=0.1 dB and selected M values.

throughput optimum. For instance, we find a margin sweet-spot around $A_{target} = 99.99\%$ where the availability A is barely degraded compared to the reference 100 G operation, and the throughput close to the maximum achievable with rate elasticity.

The results displayed in Fig. 8 clearly establish the trade-off between capacity and availability, and allow to identify a margin sweet-spot. However the *a posteriori* analysis we used is impossible to apply to a live network. Moreover, from the data analysis in II.D, there is clearly more capacity to yield when the rates are dynamically adapted than when they are statically set. In the following, we leverage the dataset to explore ready-to-deploy dynamic rate adaptation mechanisms liable to function at the margin sweet-spot identified here.

IV. Memoryless, Dynamic Rate Adaptation (M-D)

The M-D rate adaptation consists in adapting the rate as fast as possible, every 15 minutes here, solely based on the last monitored SNR value, i.e. the SNR averaged over the previous 15 minutes. We assume the rate adaptation to be instantaneous. To avoid potential rate instability should the SNR fluctuate around a threshold value, we introduce an hysteresis cycle as depicted in Fig. 9.

Let us write R_i an arbitrary rate and R_{i+1} the upper rate. The difference between R_{i+1} and R_i is the rate granularity G. As illustrated in Fig. 10, the condition to increase the rate from R_i to R_{i+1} is for the SNR to cross $T_U(R_i \rightarrow R_{i+1})$ upward. Similarly, the condition to decrease the rate from R_{i+1} to R_i is for the SNR to cross $T_D(R_{i+1} \rightarrow R_i)$ downward. We define the residual margin M and hysteresis H, both in dB scale, as:

$$\begin{aligned} M &= T_D\left(R_{i+1} \rightarrow R_i\right) - SNR_{FEC}(R_i) \\ H &= T_U\left(R_i \rightarrow R_{i+1}\right) - T_D\left(R_{i+1} \rightarrow R_i\right) \end{aligned} \tag{7}$$

where $SNR_{FEC}(R_i)$ is the FEC limit for R_i.

In the following, we study the impact of the choice of M-D parameters (H, M, G) on the network throughput T, defined in III.E, and on the *internal* availability A_{int} of the connections, defined in III.A. The latter is estimated by counting a 15-minute outage for each interval where the SNR monitored is below the FEC threshold corresponding to the rate setting. This temporarily results in $C_k = 0$, which also impacts the throughput T.

We plot in Fig. 10(a) the total network throughput achieved in M-D mode as a function of the hysteresis H. As expected, the smaller the granularity, the higher the throughput. For a given granularity, the highest throughput is achieved with a 0.01 dB hysteresis value. Although a lower hysteresis ought to increase the number of outages, this phenomenon is mitigated by a small residual margin, i.e. 0.1 dB. Also, the impact of additional outages on throughput for some connections is overbalanced by a throughput gain for the connections - a majority - not requiring hysteresis.

In Fig. 10(b), we observe that a residual margin improves the network throughput only for the smaller 1 Gb/s granularity. Indeed, with coarser granularities, the mismatch between observed performance and rate results in an intrinsic margin, similar to an unallocated margin at design [14]. Like Fig. 8 obtained with static rate settings, the bell-shape curves here denote the trade-off between capacity and availability. For each granularity, the throughput is maximized when throughput gains from higher rates balance losses from related outages.

Interestingly, the network throughput hardly depends on the residual margin above 0 dB. For deeper insight, we plot in Fig. 10(c-d) the cumulative distribution functions (CDFs) of individual *internal* availability $A_{int,k}$ and throughput T_k calculated for each connection k. Note that above four nines, the discontinuities of the *internal* availability CDF are due to the 15-minute-refresh rate of the dataset. The CDFs show that a 0.1 dB residual margin is enough to significantly improve the availability with minimal cost on the throughput. We thus identify a sweet-spot with H = M = 0.1 dB where with standard G = 25 Gb/s, throughput is increased by 115% compared to T_{ref}, with an *internal* availability above 99% for all connections, and above 99.99% for 90% of connections.

The M-D mode is interesting due to its ability to convert any temporary or permanent SNR increase into capacity, and to quickly adapt to a SNR decrease thus avoiding prolonged outages in case of soft failure. However, the applicability of the M-D mode is contingent on the transponder ability to adapt rate in a hitless manner. Although researched [19], such property is not yet a commercial reality. More, high-frequency rate adaptation may be challenging for network operation. For this reason, we explore next an opposite dynamic scheme where the rate may only be adapted after a service interruption.

V. Causal, Dynamic Rate Adaptation (C-D)

In C-D rate adaptation, we leverage the data accumulated from the last service interruption to adjust the rate after an observation period of duration T_{obs}, during which the rate is set at 100 Gb/s. Compared to the M-D mode, this could be easily deployed with today's network infrastructure and transponders.

The rate is set so that the FEC threshold is below the lower Tukey fence [20] of the SNR time series during T_{obs}, as depicted in Fig. 12. Tukey fences is a typical method to identify statistical outliers within a time series. We thus set the rate so that only a statistical outlier can lead to an outage due to insufficient SNR. The lower Tukey fence or LTF is estimated by applying the following formula:

$$LTF = Q_1 - 1.5\left(Q_3 - Q_1\right) \tag{8}$$

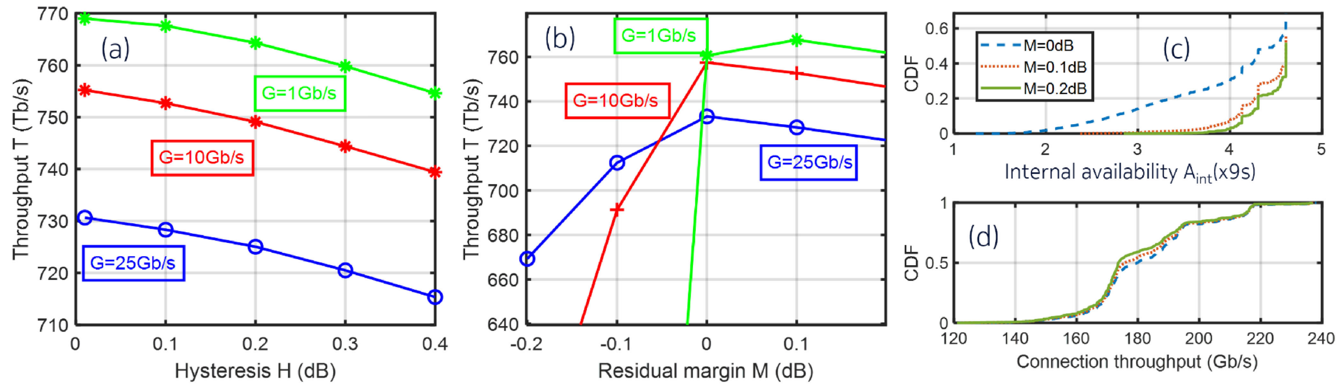

Fig. 10. Hysteresis cycle to set the rate according to the monitored SNR in M-D mode.

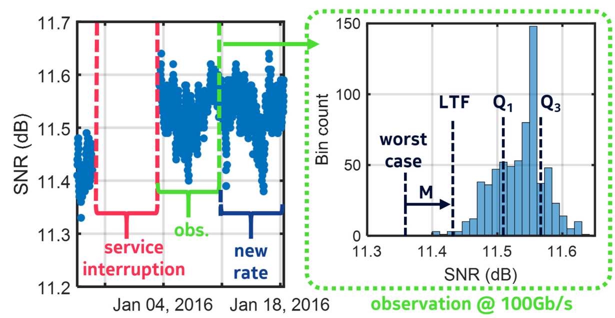

Fig. 11. Network throughput in C-D mode as a function of (a) the residual margin for T_{obs}=1 day and (b) the observation period for M=0.5 dB. CDF of connection internal availability (c) and throughput (d) for G=25 Gb/s, M=0.5 dB and selected T_{obs} values.

where Q_1 and Q_3 are respectively the first and third quartiles of the SNR distribution during T_{obs}.

To cover for potentially misleading observations during T_{obs}, we allow the possibility to add a residual margin M on top of the Tukey fence criteria. In the end, the selected rate R_{sel} verifies:

$$R_{sel} = \max\left(R : SNR_{FEC}(R) < LTF - M\right) \tag{9}$$

We study the impact of C-D parameters (M, T_{obs}, G) on the performances in terms of network throughput T and *internal* availability. Those are calculated similarly to Section IV. We first plot in Fig. 11(a) the network throughput as a function of the residual margin M for T_{obs}=1day. The impact of M is similar to what is observed with the M-D mode, i.e. a residual margin optimizes the throughput only for 1 Gb/s granularity; for larger granularities, a residual margin slightly decreases the throughput but improves the availability of all connections.

The influence of T_{obs} is investigated in Fig. 11(b-d). In Fig. 11(b), we observe that for a given granularity, the highest throughputs are achieved with the smaller observation periods. For deeper analysis, we plot in Fig. 11(c-d) the CDFs of connection *internal* availability and throughput. We observe that increasing the observation period from 6 hours to 7 days drastically reduces the throughput, but has almost no impact on availability. With SNR stationarity and Tukey fence criteria, a mere 6-hour observation period achieves 106% more throughput than T_{res} with an *internal* availability above 99.99% for more than 95% of connections.

VI. Conclusion

Through the analysis of data from a large backbone network, we demonstrated via the SNR that the QoT of a path or connection cannot be considered as Gaussian-distributed. We also quantified and discussed the various short and long term variations as well as the imbalance between the minimum and maximum values observed, mainly stemming from Q-drops. As a result, any margin reduction will impact the availability, thus the quality of service.

Although this may appear as a challenge for low-margin operation, we demonstrated with virtual, arbitrary and static rate settings that once deployed, the total throughput capacity may be largely increased - more than doubled for the studied network - with minor impact on the availability. We pointed out that a data refresh period below 15 minutes will be necessary to optimize high-availability connections, over four nines. More importantly, after quantifying the trade-off between capacity and availability, we identified a margin sweet-spot where capacity and availability are jointly optimized.

We then proposed two opposite dynamic rate adaptation mechanisms as practical solutions to reach this margin sweet-spot. For both, we observe that a 25 Gb/s granularity - the state-of-the-art for coherent transponders - is enough to convert the bulk of the margins into capacity. In comparison, the technically challenging 1 Gb/s granularity would only provide up to 10% of extra throughput.

The M-D mode we trialed achieves up to 115% throughput gain compared to default 100 Gb/s operation, while maintaining an availability over 99.99% for 90% of connections despite significant QoT variations. In comparison, the more practical and easier-to-deploy C-D mode achieves up to 106% throughput gain. The lesser gain achieved with the C-D mode is however compensated by an improved quality of service, with an availability over 99.99% for 95% of connections. Note that since both M-D and C-D schemes have pros and cons, an optimized mechanism could be found by merging the two.

Considering our simplifying assumptions and the fact that each optical network is unique, the throughput gains we achieved

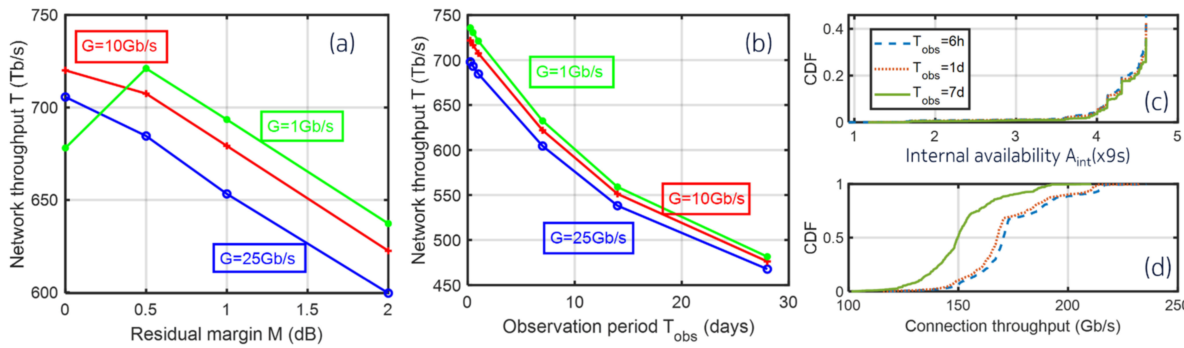

Fig. 12. Rate adaptation mechanism in C-D mode.

here should not be applicable to any network. However, considering the advent of commercial solutions for collecting monitoring data throughout optical networks and the maturity of QoT estimation models, our methodology is ready to be applied by both vendors and operators to harmlessly quantify how margin reductions would impact the availability in any network. Moreover, it could be used to identify the most favorable trade-off between capacity and availability, and the most suitable rate adaptation mechanism to achieve it.

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